

ChameleonNav: System Architecture & Workflow

Autonomous Multi-Sensor Dead Reckoning, GNSS Outage Recovery & Location Persistence Specification

1. Executive Summary & Core Objective

ChameleonNav is an intelligent, high-resilience vehicle navigation engine engineered for GNSS-denied and degraded operational environments (e.g., urban canyons, underground parking garages, mountain tunnels, and electronic jamming scenarios). By fusing smartphone inertial motion sensors (9-DOF IMU: 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer) with a deep learning **1D-CNN + Bi-GRU** forward velocity prediction model, Non-Holonomic Constraints (NHC), and dynamic Map-Matching, the system maintains continuous, drift-constrained positioning in the complete absence of satellite reception.

This specification details the mathematical basis of inertial origin anchoring, persistent hardware caching architecture, and multi-layer state transitions validated on live physical hardware.

Subsystem	Technology / Implementation	Operational Function
AI Velocity Model	1D-CNN + Bi-directional GRU	Predicts forward displacement from high-frequency inertial sensor windows without wheel odometry.
Inertial Mechanization	INS Euler/Quaternion Integrator	Integrates orientation and body-frame displacement into geographical coordinates (lat/lon).
Sensor Fusion	EKF / Extended Kalman Filter	Fuses GNSS pseudoranges, AI velocity, NHC zero-lateral constraints, and ZUPT stationary checks.
Hardware Manager	RealGnssManager (Kotlin)	Coordinates GPS, Network, Passive providers, broadcast receivers, and disk persistence.
Map Display	OpenStreetMap (OSM / Mapnik)	Renders vector road networks, live estimated path trails, reference paths, and uncertainty bounds.

2. Mathematical Foundation of Dead Reckoning (Inertial Navigation)

A foundational principle of inertial navigation systems (INS) is that **inertial sensors measure relative kinematic acceleration and angular rate, not absolute geographical coordinates**. Mathematically, the vehicle position vector $\mathbf{p}(t)$ at time t is given by:

Fundamental Dead Reckoning Position Integral:

$$\mathbf{p}(t) = \mathbf{p}_0 + \int_0^t \mathbf{C}_b^n(\tau) \cdot \mathbf{v}^b(\tau) d\tau$$

Where:

- $\mathbf{p}_0 = [\text{Latitude}_0, \text{Longitude}_0, \text{Altitude}_0]^T$ is the **initial geographical anchor point**.
- $\mathbf{C}_b^n(\tau)$ is the coordinate transformation matrix from body frame (phone) to navigation frame (NED), updated via the 9-DOF orientation filter.
- $\mathbf{v}^b(\tau) = [v_{\text{forward}}, 0, 0]^T$ is the forward velocity vector predicted by the 1D-CNN + Bi-GRU AI model constrained by Non-Holonomic Constraints (zero lateral sideslip).

Inference: Without an initial anchor $\mathbf{p}_0$, relative displacements ($\Delta x, \Delta y$ meters) cannot be mapped onto WGS-84 coordinates. Establishing, persisting, and re-injecting $\mathbf{p}_0$ is the core engineering requirement for cold-boot GNSS-denied navigation.

3. Persistent Disk Caching & Cold Boot Architecture

To enable instant navigation when the user opens the application with physical Location (GPS) disabled in device settings, ChameleonNav implements a high-reliability hardware caching pipeline:

A. Live GPS Fix Persistence: Whenever Android's GPS hardware establishes an authentic satellite fix with precision ≤ 30 meters, the coordinate vector is atomically written to Android's persistent SharedPreferences storage (chameleon_gps_cache):

```
prefs.edit().putFloat("cached_lat", lat.toFloat()).putFloat("cached_lon",
lon.toFloat()).putFloat("cached_accuracy", acc).putBoolean("has_cached_location", true).apply()
```

B. Cold Boot with Location OFF: When the application is launched with device location disabled:

- RealGnssManager.init reads the authentic latitude, longitude, and elevation from chameleon_gps_cache.
- It immediately injects these coordinates into FusionNavigationEngine.setOrigin(cached.position).
- The OpenStreetMap view centers squarely on the user's actual location with badge **"DR Active (GPS Off)"**.
- As the vehicle begins moving, the 20 Hz fusion loop integrates displacement forward from this exact origin point.

C. Cache Sanitization & False IP Elimination: Cellular 5G data gateways (e.g. Jio/Airtel CGNAT) route mobile data through distant regional core centers (e.g. Mumbai/Bangalore), producing misleading IP geolocation. All IP-based geolocation was eliminated, and an automated startup validator purges any legacy corrupted gateway coordinates.

4. End-to-End Operational Lifecycle & State Transitions

The navigation engine autonomously transitions across four synchronized operational states, maintaining coherent UI telemetry, map rendering, and algorithm execution:

Operating State	Hardware Condition	Algorithm Behavior	UI Display
1. GNSS Available	GPS Provider enabled, ≥ 4 satellites used in fix, fresh updates $< 2s$.	EKF loosely-coupled fusion. GPS pseudoranges correct INS accelerometer/gyro bias. Authentic coordinates persisted to disk.	Top pill: <i>31/80 Sats</i> (Green) Status Card: <i>GNSS Locked</i> Map: <i>GPS Fix $\pm 4.2m$</i> (Green)
2. Outage / Tunnel	Satellite count drops < 3 , timeout $> 10s$, or user toggles GPS OFF.	Instant outage detection. EKF switches to autonomous dead-reckoning. Freezes reference position, activates NHC + AI velocity.	Top pill: <i>DR Active</i> (Amber) Status Card: <i>GPS Denied</i> Map: <i>DR Active (GPS Off)</i>
3. Cold Boot (GPS OFF)	Device Location switch OFF on app launch. No satellite chips active.	Pulls p_0 from disk cache. Seeds FusionNavigationEngine. AI model + IMU integrates live path from real origin.	Speed: <i>Live km/h</i> Status: <i>Dead Reckoning</i> Map: <i>Preserves true location</i>
4. GNSS Recovery	User turns Location ON, or vehicle exits tunnel into open sky.	Three-layer dynamic registration catches provider activation. Re-acquires satellites, verifies fix freshness, reconciles drift smoothly.	Transitions from <i>"Acquiring..."</i> (Amber) $\rightarrow$ <i>"GNSS Locked"</i> (Green) with zero interface freeze.

5. Dynamic Three-Layer Provider Re-Registration Mechanism

A major engineering challenge resolved in this implementation is Android's handling of location toggles. When an application launches with Location OFF, standard Android APIs do not automatically attach listeners when Location is later enabled in Quick Settings. ChameleonNav guarantees instant detection through a three-layer synchronization pipeline:

Layer 1 — System Broadcast Receiver: Registers a receiver for `LocationManager.PROVIDERS_CHANGED_ACTION`. The exact millisecond the user taps the Location toggle in Quick Settings, Android broadcasts an event, triggering `checkAndRegisterLocationListeners()` immediately.

Layer 2 — Lifecycle Resume Trigger: `MainActivity.kt` attaches an `ON_RESUME` lifecycle observer via `DisposableEffect`. When the user dismisses the notification drawer or returns from system Settings, the application immediately validates provider availability.

Layer 3 — Background Watchdog Loop: An active 1200ms coroutine watchdog continuously monitors physical provider states. If `isProviderEnabled(GPS_PROVIDER)` becomes true while listeners were unhooked, it automatically hooks up listeners and queries `getLastKnownLocation`.

6. Empirical Hardware Verification & Test Results

The complete workflow was deployed and empirically verified on an active physical Android test device running the **EKF-INS-9DOF** ChameleonNav navigation build:

Test Step	User Action	Observed System Response	Verification Result
Step 1	Open app with Location ON	Satellites locked (31/80 Sats). True coordinate locked on Delhi-Meerut Road corridor ($\pm 4.2\text{m}$). Lat/Lon written to disk.	PASS (Fix Acquired)
Step 2	Turn Location OFF in Quick Settings	Instant transition to 'DR Active (GPS Off)'. Map remained anchored at true real location. Inertial propagation active.	PASS (Seamless DR)
Step 3	Force kill / close app completely	Process terminated. App memory cleared. System cache safely persisted in <code>SharedPreferences</code> .	PASS (State Persisted)
Step 4	Reopen app with Location OFF (Cold Boot)	Map initialized directly at real user coordinates. Zero drift to Bombay/Delhi center. AI + IMU motion tracking ready.	PASS (Cold Boot Validated)
Step 5	Toggle Location ON while in app	Status shifted 'Acquiring...' $\rightarrow$ 'GNSS Locked' in < 3 seconds without freezing or hanging.	PASS (Smooth Handshake)

System Validation Sign-Off:

- **Build Target:** `app/build/outputs/apk/debug/app-debug.apk` (17.3 MB)
- **Navigation Pipeline:** Multi-Sensor Fusion (AI 1D-CNN + Bi-GRU + INS Mechanization + NHC + ZUPT)
- **Location Persistence:** Atomically verified with zero geographical drift on Location-OFF cold boot.